

State-of-Charge Equalizations Comparison in a Multi-prosumer Environment Based on Real-Time Energy Management Systems with IoT-Support

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Abstract—One essential distributed energy resource (DER) in smart microgrids (MGs) is the Energy Storage System (ESS) based on batteries, which enables many power and energy functionalities. However, when a microgrid includes more than one ESS, or when multiple ESSs interact within a multi-prosumer environment, their charging and discharging processes should be managed to avoid uneven degradation. This paper presents the modeling and implementation of two different strategies for scheduling the DERs using mixed-integer linear programming (MILP) and quadratic programming (QP) in a multi-prosumer grid-connected microgrid. Each prosumer has its own battery-based ESS with various capacities. The proposed strategies were implemented as a real-time energy management system (EMS) in an experimental laboratory Microgrid based on a rolling horizon scheme. The equalization of the ESSs' state of charge (SoC) is achieved in both cases but there is a compromise between cost and execution time. Meanwhile, the IoT-based smart meter allows the EMS to react in real-time to mismatches between predicted and real load.

Keywords— Economic Dispatch problem, rolling horizon, Optimization, state of charge, internet of things, microgrid testbed

I. INTRODUCTION

MG concept appears as a systematic approach for dealing with the Smart Grids, as a well-defined energy entity integrating DERs, distributed generators, and loads that include proper controls and management that enable their islanded or grid-connected operation [1]–[3], [14]. MGs based on multiple prosumers imply the interaction of various battery-based ESS [4]. Within this interaction, differences between cycles of charge and discharge may cause uneven degradation of the storage devices [5]. To reach an efficient utilization of the

ESSs and prolong their service life, it is better to balance or equalize ESS SOC of the prosumers [6].

SoC equalization problem between distributed ESS has been typically treated as a non-optimal problem in which the main considerations are the current SoC status of the distributed batteries and the current power balance [5]. Therefore, the equalization strategy just adjusts the power contribution of each ESS unit. This may cause power to flow between energy storage devices which in the end is reflected in more power losses [5]. Also, it is important to say that the proposed SoC equalization methods barely consider the whole operation of the power system. This paper includes the SoC equalization within the optimal operation schedule for the whole power system. In this way, equalization not only takes part in the even operation of the energy storage but also takes an active part within the optimal operation of the whole Multi-prosumer environment. Here, the equalization is boarded from two approaches one linear and one quadratic showing different performances between operation cost and processing time.

This paper presents the development and experimental test of an EMS for MG with the ability to equalize ESSs' SoC while integrating the monitoring of measurements collected by smart meters (SM) based on Internet of Things (IoT) technology, a rolling horizon scheme, and a scheduling module. The specific study case is a grid-connected MG with three prosumers, denoted as inverter-based distributed generators (IDG), as shown in Fig. I. Here, each prosumer is composed of a photovoltaic system (PV), an ESS, and a local load. The multi-prosumer microgrid also feeds two common loads, a shed constant load, measured by an SM, and a variable load that represents the forecasted consumption. In this way, it is possible to emulate the differences between prediction and real operation. Two optimization problems have been modeled as MILP and QP by using MATLAB and experimentally compared. The problems aim to reduce the operational cost and the error between the SoCs.

This work was supported by Research and Development Department, Sonnen Inc., Atlanta, GA 30033 USA (e-mail: e.sanabria@sonnen-batterie.com) and Minciencias with the project "Programa de Investigación en Tecnologías Emergentes para Microredes Eléctricas Inteligentes con Alta Penetración de Energías Renovables" [grant number 80740-542-2020]. The views presented are the opinions of the authors and do not represent any official position of the sponsoring organization

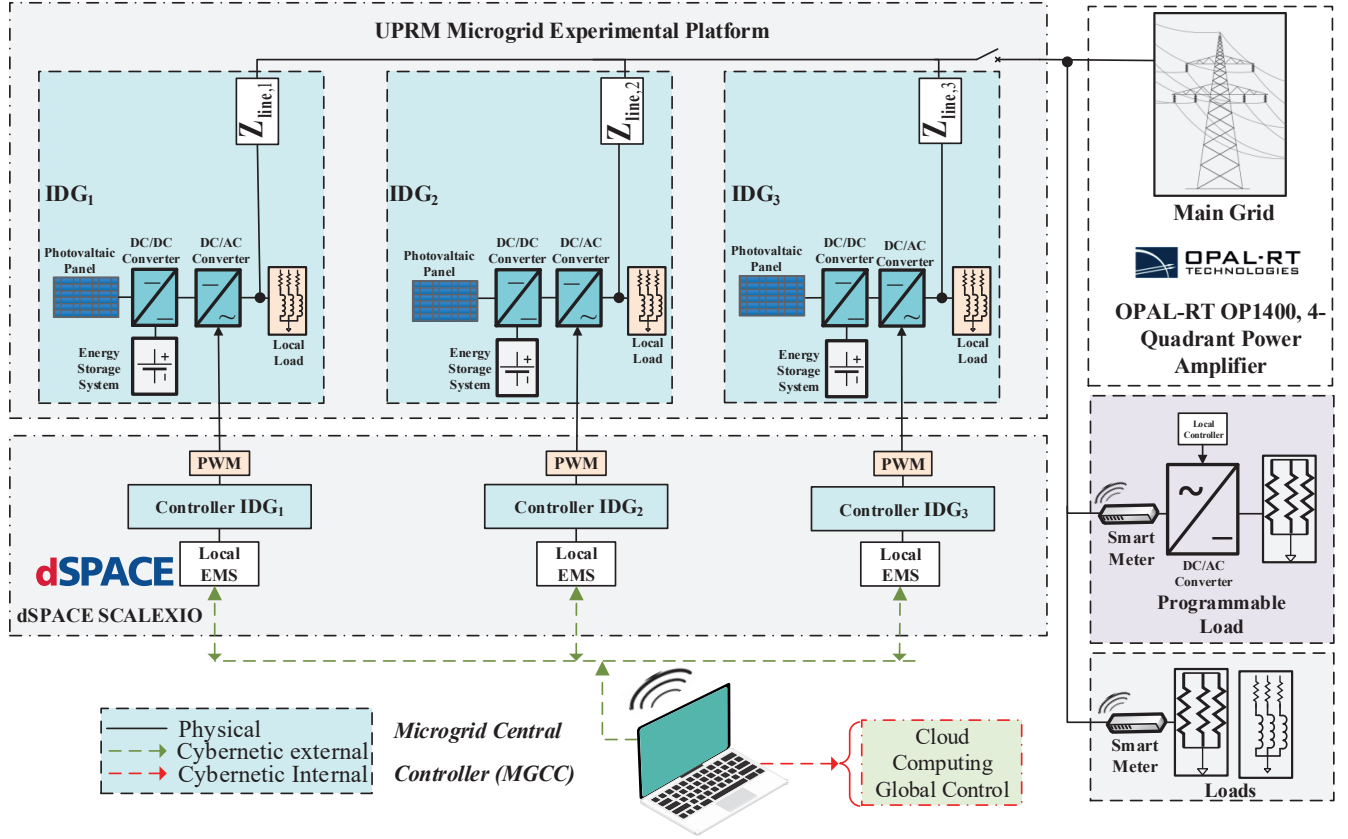


Fig. 1. Diagram of the experimental platform.

II. ENERGY MANAGEMENT SYSTEM

The main functionality of EMS is the scheduler, which has been developed through two defined optimization problem taking into account measurements, technical constraints and predicted data. To cope with the associated uncertainties related to the variable nature of the case study, the EMS has been implemented in a rolling horizon scheme.

Considering an Economic Dispatch Problem mathematically modelled solution [5], previous research formulate the optimization problem using indexes like $i \in N$ for distributed units and $t \in T$ for time as in [4]. This paper defines formulation using these indexes where i represents the prosumers and t each time spam, which lasts Δt .

For the equalization of the SoC of the ESSs, two optimization problems have been modeled as MILP and QP by using MATLAB and then experimentally compared [7]. They differ in the definition of the objective function and in some constraints, which have been defined as follows based on previous research [8], [9].

The optimization problems aim at minimizing the cost of using energy from the grid and the prosumers, while includes a penalty that seeks to minimize the error among the SoC of the ESSs, $E_{SoCi,t}$. This penalty leads to the equalization of

the SoCs. Thus, the objective function can be written as,

$$\min_x \sum_{i \in N} \sum_{t \in T} C_{i,t} \cdot p_{i,t} + C_{i,t} \cdot p_{Grid,t} + C_{Ei,t} \cdot E_{SoCi,t} \quad (1)$$

where $C_{i,t}$ is the tariff of the energy of the grid and $C_{Ei,t}$ is the penalty coefficient of the Error of SoC among ESSs. The difference between the objective function of the proposed models is the definition of $E_{SoCi,t}$, one is lineal and the other is quadratic.

The first common constraint defines the energy balance as,

$$\sum_{i \in N} p_{i,t} \Delta t + p_{Grid,t} \Delta t = p_{d,t}^{forecast} \Delta t + p_{d,t}^{smart} \Delta t = 0, \quad \forall t \quad (2)$$

where $p_{i,t} \Delta t$ is the energy provided by the i -th prosumer, $p_{Grid,t} \Delta t$ is the energy exchanged with the grid, and the energy requested to the load is represented by $p_{d,t}^{forecast} \Delta t + p_{d,t}^{smart} \Delta t$. In turn, the energy provided by the prosumers comes from their PVs and ESSs and can be written as,

$$p_{i,t} \Delta t = p_{pv,i,t} \Delta t + (p_{Batt,disch,i,t} \Delta t - p_{Batt,charg,i,t} \Delta t) = 0, \quad \forall i, t \quad (3)$$

where $p_{pv,i,t} \Delta t$ is the forecast PV energy, while the battery energy has been modeled with two variables $p_{Batt,disch,i,t} \Delta t$ and

$p_{Batt,charg_{i,t}} \Delta t$, which represent the discharged and charged energy, respectively.

The relationship between these ESS energy variables and the SoC can be defined as,

$$SoC_{i,t} = \begin{cases} SoC_{i,0} - \frac{p_{Batt,disch_{i,t}} \Delta t \cdot 100\%}{\eta_{disch_i} \cdot C_i^{Batt}} + \frac{\eta_{charg_i} p_{Batt,charg_{i,t}} \Delta t \cdot 100\%}{C_i^{Batt}}, & \forall i, t = 1 \\ SoC_{i,t-1} - \frac{p_{Batt,disch_{i,t}} \Delta t \cdot 100\%}{\eta_{disch_i} \cdot C_i^{Batt}} + \frac{\eta_{charg_i} p_{Batt,charg_{i,t}} \Delta t \cdot 100\%}{C_i^{Batt}}, & \forall i, 1 \leq t \leq T \end{cases} \quad (4)$$

where $SoC_{i,0}$ are the initial conditions of the SoCs, C_i^{Batt} are the capacities in kWh of the ESSs, whereas η_{charg_i} and η_{disch_i} are the charging and discharging efficiencies, respectively.

The common constraints that define the boundaries of the variables have been defined as,

$$p_{min,i,t} \leq p_{i,t} \leq p_{max,i,t}, \quad \forall i, t \quad (5)$$

$$0 \leq p_{pv_{i,t}} \leq p_{pv_{nom_i}}, \quad \forall i, t \quad (6)$$

$$0 \leq p_{Grid,t} \leq p_{Grid_{max}}, \quad \forall t \quad (7)$$

$$SoC_{i,t}^{min} \leq SoC_{i,t} \leq SoC_{i,t}^{max}, \quad \forall i, t \quad (8)$$

The limits in the generation level are defined by (5), (6) and (7) according to the power boundaries of the prosumers, considering PVs and ESSs, the forecast power generation, and the utility contribution, respectively. Meanwhile, the SoC of the ESSs should be the range defined in (8), which is usually recommended by the manufacturers.

A. Linear Model

For the mathematical formulation of the MILP, the error $E_{SoC_{i,t}}$ has been written as,

$$E_{SoC_{i,t}} = SoC_{i,t} - SoC_{i+1,t}, \quad \forall i, t \quad (9)$$

and the penalty coefficient has been selected so that equalization has the lowest priority of minimization.

To hold the linearity of the model, the errors among the SoCs must be positive, which have been defined as,

$$SoC_{i,t} - SoC_{i+1,t} \leq 0, \quad \forall i, t \quad (10)$$

For this reason, a decision-making engine has been defined to pre-order the prosumer's generation entries based on the layout that allows positive errors as shown in Fig. II-A. To illustrate, having three prosumers and for an specific time j , three errors are considered:

$$E_{SoC1} = SoC_1 - SoC_2, \quad (11)$$

$$E_{SoC2} = SoC_2 - SoC_3, \quad (12)$$

$$E_{SoC3} = SoC_3 - SoC_1 \quad (13)$$

If $SoC_1 < SoC_2 < SoC_3$, E_{SoC1} and E_{SoC2} would be negative. The decision-making engine previously changes the

order of the SoCs and guarantee the errors are positive by defining

$$E_{SoC1} = SoC_{k11} - SoC_{k12} = SoC_2 - SoC_1, \quad (14)$$

$$E_{SoC2} = SoC_{k21} - SoC_{k22} = SoC_3 - SoC_2, \quad (15)$$

$$E_{SoC3} = SoC_{k31} - SoC_{k32} = SoC_3 - SoC_1 \quad (16)$$

Additionally, the model includes some constraints to define a feasible region. In this case, the binary variable $x_{batt,t}$ is used to define the status of the charging process, i.e., its value is 1 when the ESS is been charged and 0 when is been discharged. This variable is used to define the upper boundaries of the power variables associated with the charge and discharge of the ESSs, which has been defined as,

$$0 \leq p_{Batt,chg_{i,t}} \leq p_{Batt,chg_{i,t}} \cdot x_{batt,t}, \quad \forall i, t \quad (17)$$

$$0 \leq p_{Batt,disch_{i,t}} \leq p_{Batt,disch_{i,t}} \cdot (1 - x_{batt,t}), \quad \forall i, t \quad (18)$$

B. Quadratic Model

As in the previous model, this optimization problem has been defined to minimize the cost of using energy from the grid and the IDGs and includes a penalty that allows the SoC equalization of the ESSs. The difference in the objective function lies in that the error among the SoCs, $E_{SoC_{i,t}}$, is squared [8], and has been written as,

$$E_{SoC_{i,t}} = (SoC_{i,t} - SoC_{i+1,t})^2, \quad \forall i, t \quad (19)$$

In this way, the error can be positive or negative, and then, the decision-making engine of the previous approach is not required.

The constraints defined above are kept, except for the boundaries of the errors among SoC (10) and the definition of the binary variable, which is not included.

In this way, the boundaries of the charging and discharging powers associated with the ESSs are defined as,

$$0 \leq p_{Batt,chg_{i,t}} \leq p_{Batt,chg_{i,t}}, \quad \forall i, t \quad (20)$$

$$0 \leq p_{Batt,disch_{i,t}} \leq p_{Batt,disch_{i,t}}, \quad \forall i, t \quad (21)$$

III. EXPERIMENTAL VALIDATION PLATFORM

The proposed EMS has been implemented on a scalable and autonomous cloud-computing experimental platform as presented in [9]. The cloud-based EMS manages an experimental microgrid platform at the Sustainable Energy Center (SEC) of the University of Puerto Rico at Mayagüez [10]. Fig. 3. shows the laboratory setup that includes four inverters with LCL output filters connected to the main grids and loads, LEM sensors and smart meters, a real-time platform, and a microgrid central computer [9]. For this development, the rolling horizon EMSs, which can deal with the mismatch between the forecast profiles and the real performance [12], were implemented on a cloud server with MATLAB to solve the optimization problems based on forecasted data, measurements of the electrical system as in [13], and the IoT data from the Smart-meters. Finally, the EMS sends the set points to the inverters at the Real-time Platform that interconnects the prosumer to the power in accordance with the optimized solution.

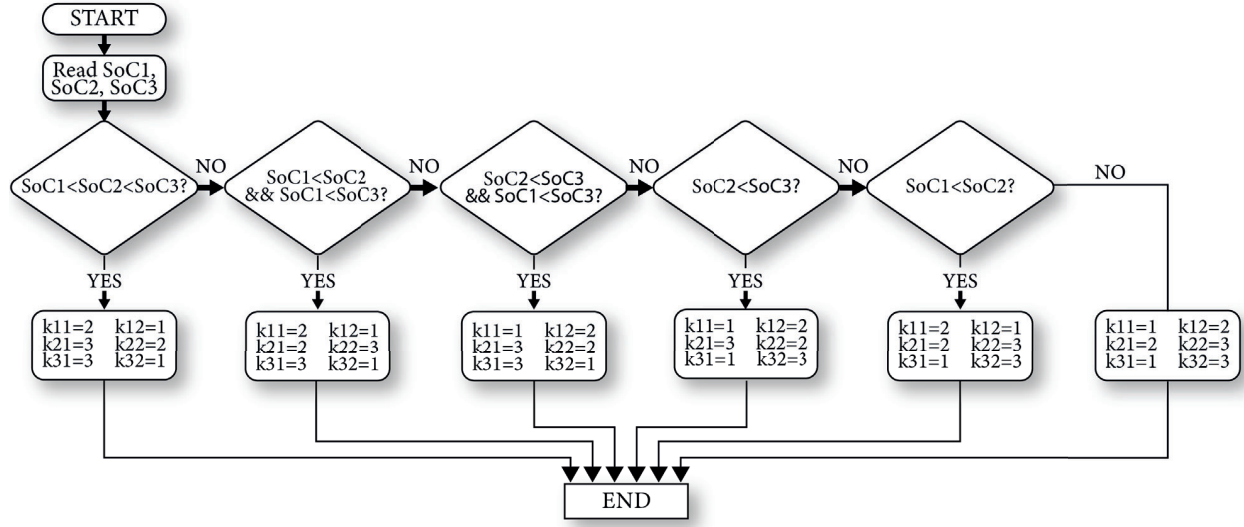


Fig. 2. Flow diagram of SoC constraints for the Linear cost function.

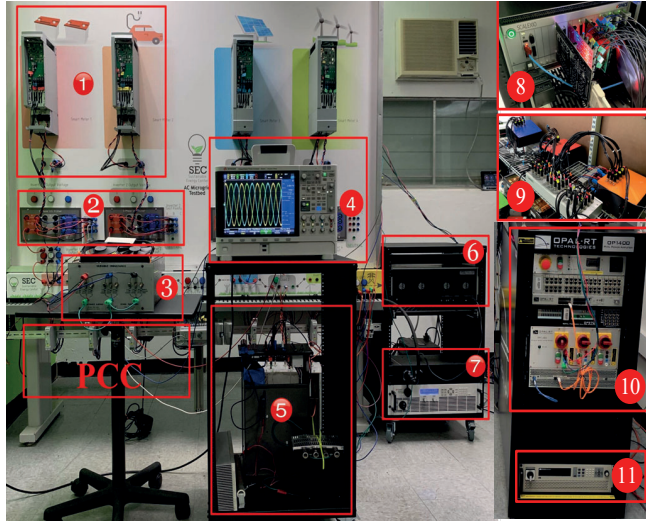


Fig. 3. Microgrid Testbed [9], [11], 1-IDGs, 2-LCL output filters, 3-Linear Load, 4-Power analyzer, 5-Non-Linear load, 6-dSPACE-Scalexio (back view), 7-DC Supply (IDGs), 8-dSPACE-Scalexio (front view), 9-Sensor Boxes, 10-OP1400 Power Amplifier, 11-DC Supply (OP1400).

IV. EXPERIMENTAL RESULTS

To conduct the experimental tests the PV generation profile was taken from a week in the summer of July 2021 from the meteorological station of the SEC (Fig. 4). The variable load profile was taken from the residential devices used daily in three different houses, using smart metering, the profile represents the use of the three houses for one week (Fig. 5). The average energy value of all houses is 45.46kWh/day. The time step of the profile is 30 min.

For the experimental verification, the window of the rolling horizon was defined as one week and it was repeated each 15 min. Figures 6 and 7 show the scheduling obtained with the

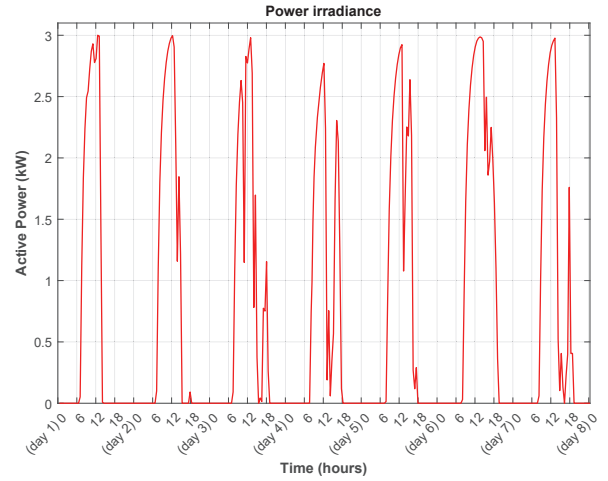


Fig. 4. Generation profile, the profile is taken from a week in the summer of July 2021 from the meteorological station of the sustainable energy center of the University of Puerto Rico, the installed capacity of PVs in each unit is 3kw, and the time steps of the profile is 30 min, that means 336 points for a week.

quadratic model and the linear model, respectively. In both cases, a delay is observed since the smart meters lasted to send new data about 4 min and, the EMS can have some non-update data. Figures 6 and 7 also highlight the difference between forecasted and real load. Thanks to the rolling horizon approach, the EMS can deal with the mismatches and recalculate the new optima set points.

Additionally, it is possible to see that with the quadratic model, the main grid supplied more energy than with the linear approach, especially during the first day. In the zoomed box is shown that also the energy injected to the main grid is higher.

Table I presents a performance comparison of cost function. Related to the energy consumption form the main grid, the

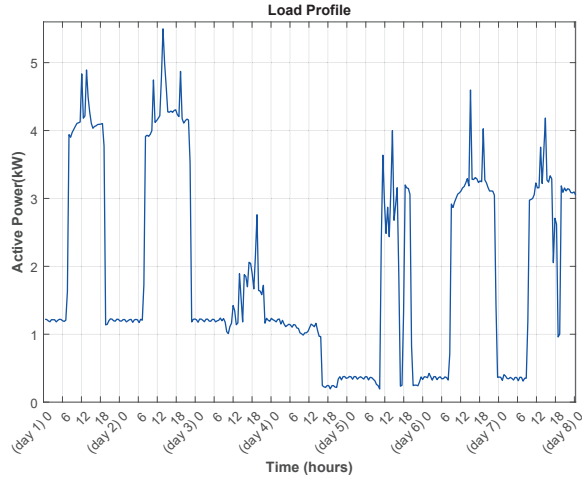


Fig. 5. Total load profile, the profile is taken from the residential devices used daily in three different houses, using smart metering, the profile represents the use of the three houses for one week. The average energy value of all houses is 45.46kWh/day, the load profile time steps are 30 min, which means 336 points per week.

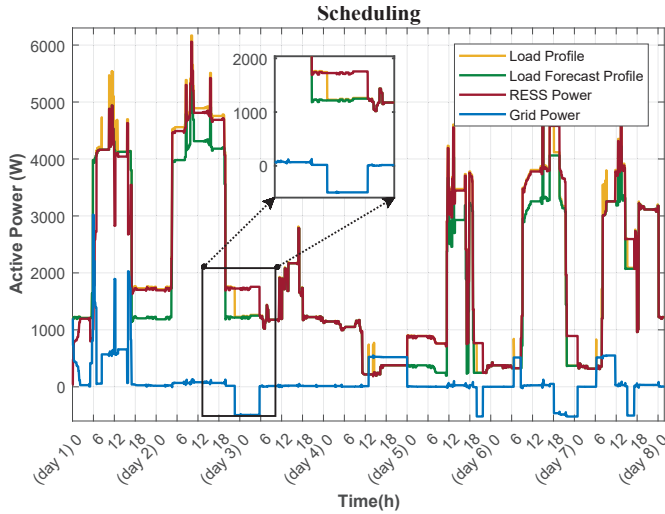


Fig. 6. Power profile of the system quadratic.

linear approach was a better performance, while the time execution was slightly higher with the linear approach

Meanwhile, in Figures 8 and 9 are shown, the SoC performances for the different storage devices for the quadratic and linear model respectively. The equalization is achieved with both approaches with almost no difference in performance and equalization time. Just a slightly better performance in terms of equalization time can be seen for the quadratic model.

V. CONCLUSIONS

This paper presents two optimization problems to minimize the cost and achieve equalization among ESSs based on batteries. The proposed approaches were able to perform equalization, with slightly better results using the quadratic

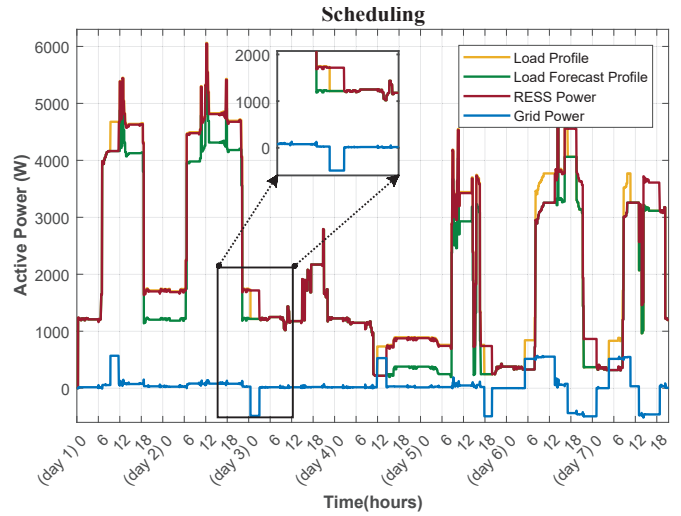


Fig. 7. Power profile of the system linear.

TABLE I
PERFORMANCE COMPARISON TABLE OF COST FUNCTIONS

Paramter	Value
Energy consumption from the utility grid by the linear cost function	2.33 kWh/week
Energy consumption from the utility grid by the quadratic cost function	5.36 kWh/week
Time execution of the linear cost function algorithm	2.01 h
Time execution of the quadratic cost function algorithm	1.84 h

model. Both approaches avoid deep discharge at the battery-based ESS, and, thanks to the equalization, uneven degradation is prevented. However, the approaches exhibit very different performance related to the energy consumption, in which with the linear approach, the MG spend about half the cost compared to quadratic. In this sense, it should be noted that prioritizing equalization can greatly affect the cost. Finally, the execution time is less with the quadratic model due to the processing time consumed by the decision-making engine together with the performance of the solver that deals with the binary variables. The real time measurement obtained from the IoT-based Smart meters, enables the rolling horizon EMS to assure the optimal operation even under load variation.

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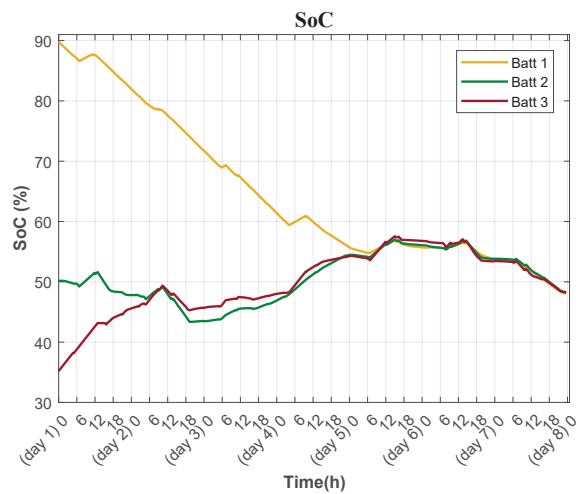


Fig. 8. SoCs behavior with quadratic model.

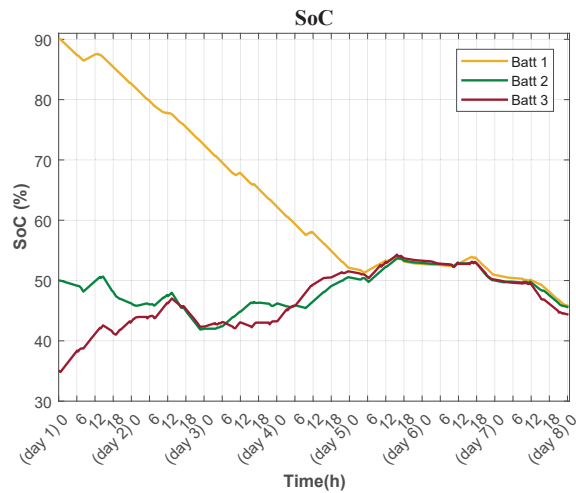


Fig. 9. SoCs behavior with linear model.

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